BorNEO HackWknd 2026 Report

Case Study 7: AI for Financial Literacy

DriveWise AI
Smart Car Finance & Wealth Impact Simulator

Prepared by:
Imperium from Universiti Putra Malaysia (UPM)

No.	GROUP MEMBERS
1	MOHAMMAD ARRAZAQ ZIKRY BIN MOHD AZHAR
2	ZAKARIA BIN ALI
3	NOR IRFAN BIN NORZAIMI
4	SRIRAM A/L SUNDRADAS
5	MUHAMMAD HAKIM ASYRAF BIN ISMAIL

Table of Contents

1. Introduction 3
    1.1. Background of Case Study 3
    1.2. Chosen SDG 3
    1.3. Problem Statement 4
    1.4. Objectives 4
2. Review of Existing Solutions 5
3. Methodology 6
    3.1. Technical Track 6
    3.2. Architecture 6
    3.3. Non-Technical Track 7
4. Results 7
5. Prototype Showcase 8
6. AI acknowledgement 17
7. Conclusion 17
8. References 18

1. Introduction

1.1. Background of Case Study

Purchasing a car is one of the most significant financial decisions for young adults and fresh graduates. In many ASEAN countries, including Malaysia, buyers frequently evaluate affordability based primarily on monthly loan installments, without fully considering total ownership costs (OECD, 2021). However, vehicle ownership involves additional recurring expenses, such as insurance, fuel, maintenance, and road tax, which substantially increase the total financial commitment (Stango & Zinman, 2009; Investopedia, 2023).

Research shows that low financial literacy often leads individuals to underestimate these hidden costs, resulting in overcommitting financially and higher long-term debt risk (Lusardi & Mitchell, 2014; World Bank, 2022). Existing financial tools generally focus on general budgeting or loan calculations and rarely provide purchase-specific insights that reflect realistic ownership costs (Consumer Financial Protection Bureau, 2016).

To address this gap, **DriveWise AI** introduces a smart affordability advisory platform that simulates realistic car ownership costs and provides actionable financial insights, empowering users to make informed and responsible vehicle purchase decisions.

1.2. Chosen SDG

This project aligns with United Nations Sustainable Development Goal (SDG) 8: Decent Work and Economic Growth. SDG 8 promotes sustainable economic growth and financial stability. Financial literacy and responsible financial planning play a crucial role in supporting individuals to make informed financial decisions.

DriveWise AI contributes to this goal by helping young adults evaluate the financial impact of car ownership, understand the true cost of vehicle ownership, and avoid unsustainable financial commitments. By encouraging smarter financial decisions, the platform supports long-term financial wellbeing and economic resilience.

Additionally, the project indirectly supports SDG 9: Industry, Innovation, and Infrastructure, by leveraging digital technology and artificial intelligence to improve consumer decision-making.

1.3. Problem Statement

Across ASEAN, digital financial services are expanding rapidly, with digital payments in the region projected to exceed USD 1.5 trillion by 2030. However, financial literacy levels have not progressed at the same pace. Studies indicate that a significant portion of adults in ASEAN lack sufficient financial knowledge to make informed financial decisions, leaving many vulnerable when managing loans, credit, and large purchases (OECD, 2021; World Bank, 2022).

One of the most common major financial commitments made by young adults and working professionals is the purchase of a car. However, many buyers evaluate affordability primarily based on monthly loan installments, without fully understanding the total cost of vehicle ownership. In reality, car ownership involves additional recurring expenses such as fuel, insurance, maintenance, and road tax, which significantly increase the long-term financial burden.

Research in consumer finance shows that individuals frequently focus on monthly payment amounts rather than the total loan cost, which can lead to suboptimal borrowing decisions and higher financial risk (Stango & Zinman, 2009; Lusardi & Mitchell, 2014). Furthermore, reports on auto financing highlight that many consumers struggle to fully understand loan terms and financing structures, increasing the likelihood of financially unsustainable commitments (Consumer Financial Protection Bureau, 2016).

Despite the financial significance of vehicle ownership, existing financial tools often focus on general budgeting or expense tracking and rarely provide decision-specific guidance for major purchases such as cars. As a result, many individuals lack accessible tools that simulate realistic ownership costs and help them understand the long-term financial consequences of their purchasing decisions

1.4. Objectives

The objectives of this project are:

1. To develop a smart affordability advisory system that evaluates whether a car purchase is financially sustainable.
2. To simulate the full cost of car ownership including loan payments, fuel, and maintenance.
3. To provide users with personalized car budget recommendations based on their financial profile.
4. To visualize financial risk through affordability scores and stress simulations.
5. To assist users in locating nearby car dealerships that match their recommended budget.

2. Review of Existing Solutions

A number of financial planning and automotive platforms currently exist, yet most provide limited guidance on the true affordability of car ownership. Traditional banking applications and online financial calculators generally estimate loan installments based on vehicle price, interest rates, and tenure. While useful for basic financing calculations, these tools fail to account for additional recurring and hidden costs, such as fuel consumption, insurance premiums, maintenance, and road taxes, which significantly affect the total financial burden of vehicle ownership (Stango & Zinman, 2009; Lusardi & Mitchell, 2014).

Similarly, car marketplace platforms, including dealership websites, focus primarily on promoting vehicle sales rather than providing financially informed guidance to consumers. These platforms are structured for marketing purposes, not for educating buyers on long-term financial implications.

As a result, many individuals continue to rely on estimates or incomplete information when making major vehicle purchase decisions, which increases the likelihood of financial overcommitment and stress. **DriveWise AI** addresses these limitations by integrating

affordability simulation, financial health scoring, and personalized dealership recommendations into a single, user-friendly platform. This approach combines practical financial education with actionable purchasing guidance, helping users make informed, responsible, and financially sustainable decisions.

3. Methodology

3.1. Technical Track

The prototype was developed using a modern web development stack optimized for fast prototyping and responsive user interfaces.

Technology Stack:

- React.js – Frontend framework for building dynamic, interactive user interfaces.
- Vite – Build tool for efficient development and fast bundling of the application.
- Tailwind CSS – Utility-first CSS framework for responsive design and consistent styling.
- Lucide Icons – Icon library to enhance user interface clarity and visual appeal.
- Google Maps Embed API – Displays nearby car dealerships for personalized recommendations.

3.2. Architecture

The system follows a frontend-only architecture, with all calculations and simulations performed in the browser. User inputs are temporarily stored using localStorage to maintain session state during navigation.

The application is structured into five main interface modules:

1. Home Page – Introduction and overview of the platform.
2. Financial Profile Input Page – Users enter monthly salary, savings, expenses, commitments, and desired car details.

3. Affordability Analysis Dashboard – Displays debt-to-income, emergency fund impact, financial health score, stress simulations and recommendations cars.
4. Car Selection Page - Displays the annual and monthly cost, Lifestyle Impact meter, Financial Safety Score, Loan Details and Life Impact Simulator.
5. Dealership Locator – Shows nearby car dealerships matching the user's recommended budget.
6. Summary Screen - show overall analysis review summary with recommended cars along with the financial scores and recommended monthly payments.

Affordability analysis is based on rule-based calculations following financial best practices, such as limiting transportation expenses to a sustainable percentage of disposable income.

3.3. Non-Technical Track

For non-technical submissions, the platform design followed a simplified Design Thinking process consisting of five stages:

1. Empathize – Understanding the financial challenges faced by young car buyers who often lack planning tools.
2. Define – Identifying the core problem: the absence of accessible tools to simulate realistic car ownership costs.
3. Ideate – Brainstorming potential solutions including affordability scoring, stress simulation dashboards, and dealership locators.
4. Prototype – Developing a functional frontend prototype that demonstrates the complete user journey and financial insights.
5. Test – Evaluating usability and clarity, ensuring the platform communicates financial information effectively to users.

This methodology ensures the platform is both practical and educational, demonstrating financial literacy tools in a format accessible to ASEAN students.

4. Results

The prototype successfully demonstrates the concept of an AI-assisted car affordability advisory system. Users can input their financial information, including salary, savings, monthly commitments, and desired car price.

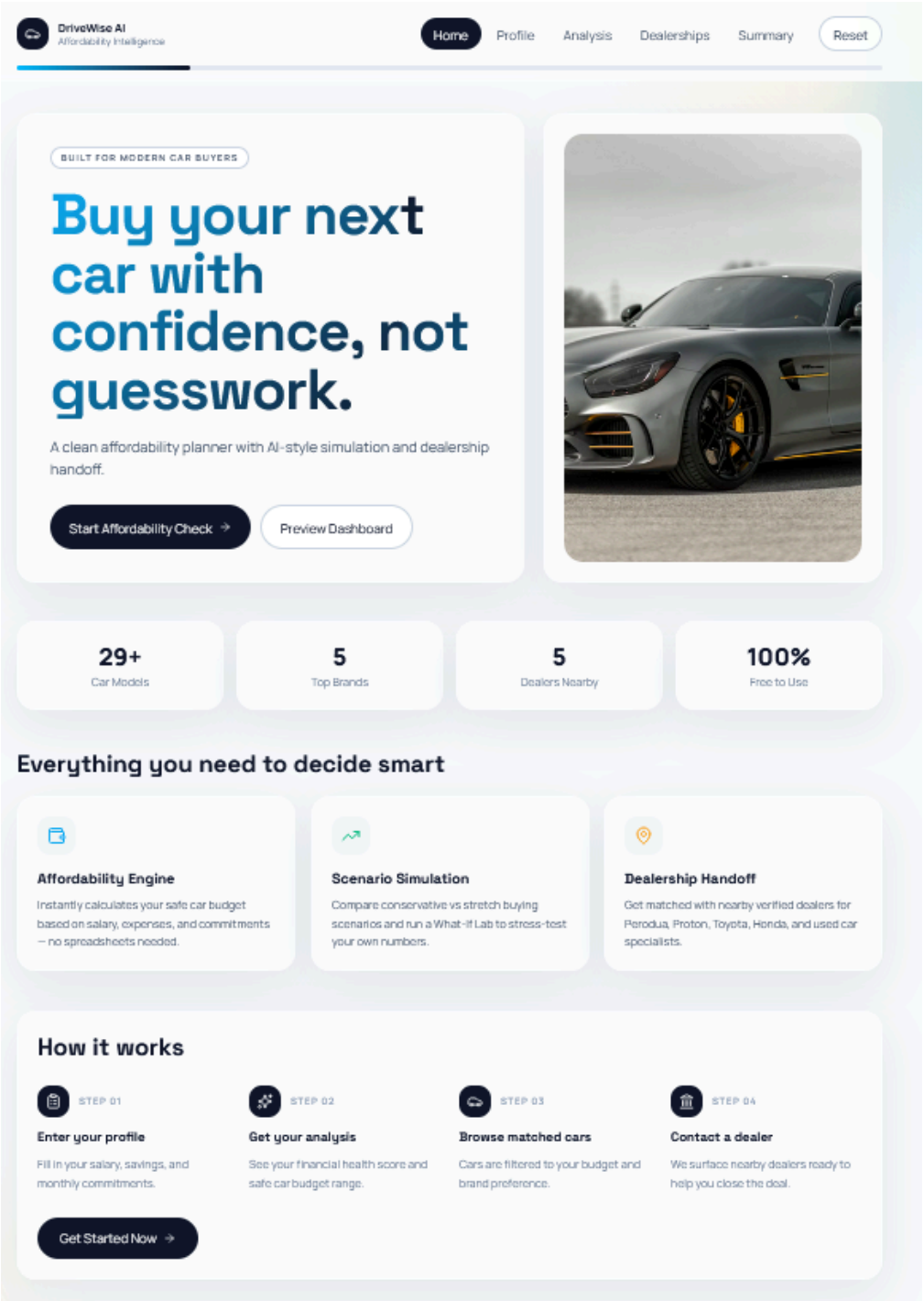
Based on these inputs, the system provides:

- Affordability analysis with financial health scores, safe car budgets, and monthly stress simulations.
- Personalized recommendations to guide users toward financially responsible car purchases.
- Actionable next steps, including nearby dealership locations aligned with the user's recommended budget.

These results indicate that the platform effectively achieves its objectives by combining financial insights, interactive simulations, and practical guidance, promoting informed and sustainable decision-making for young car buyers.

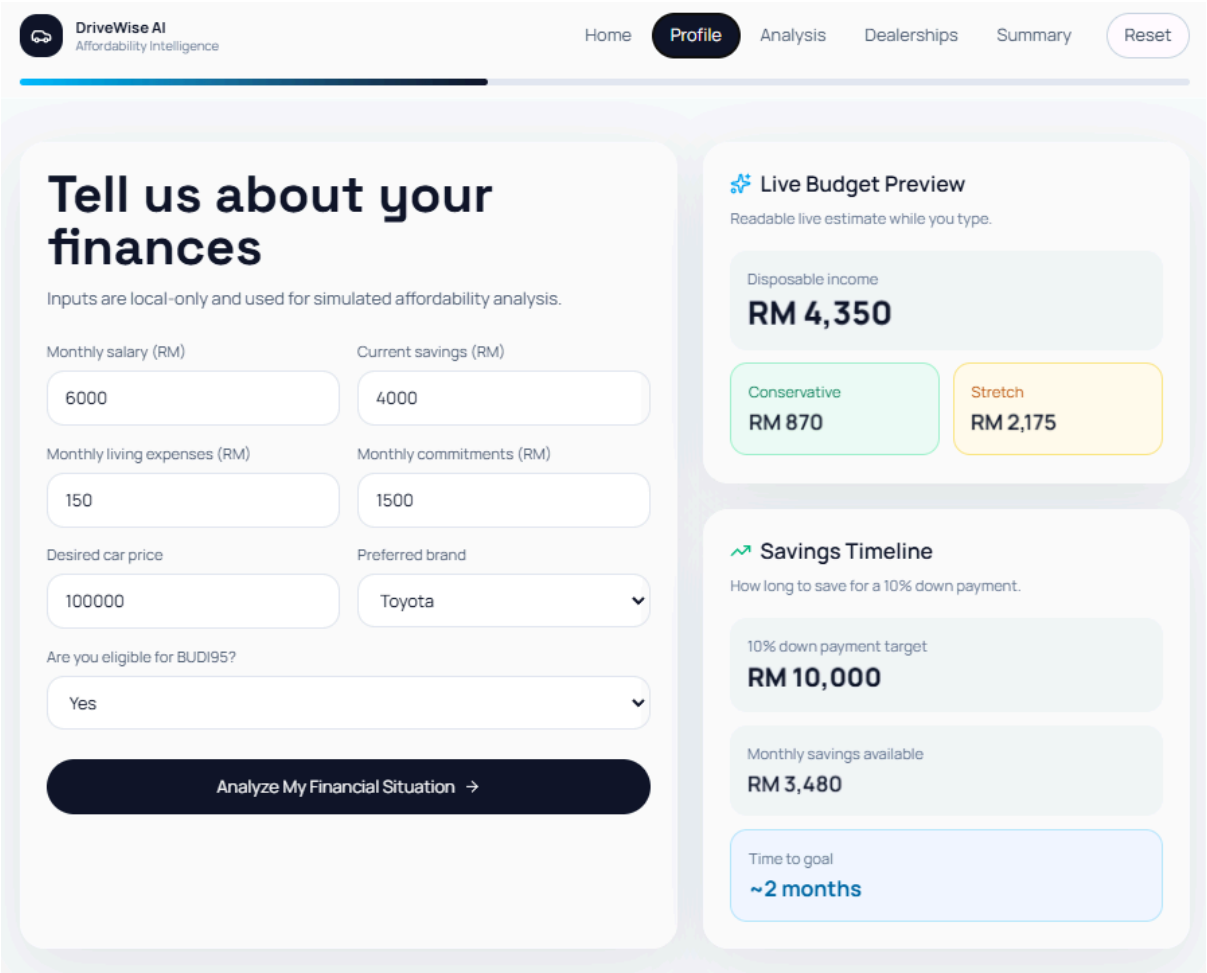
5. Prototype Showcase

I . Home Interface



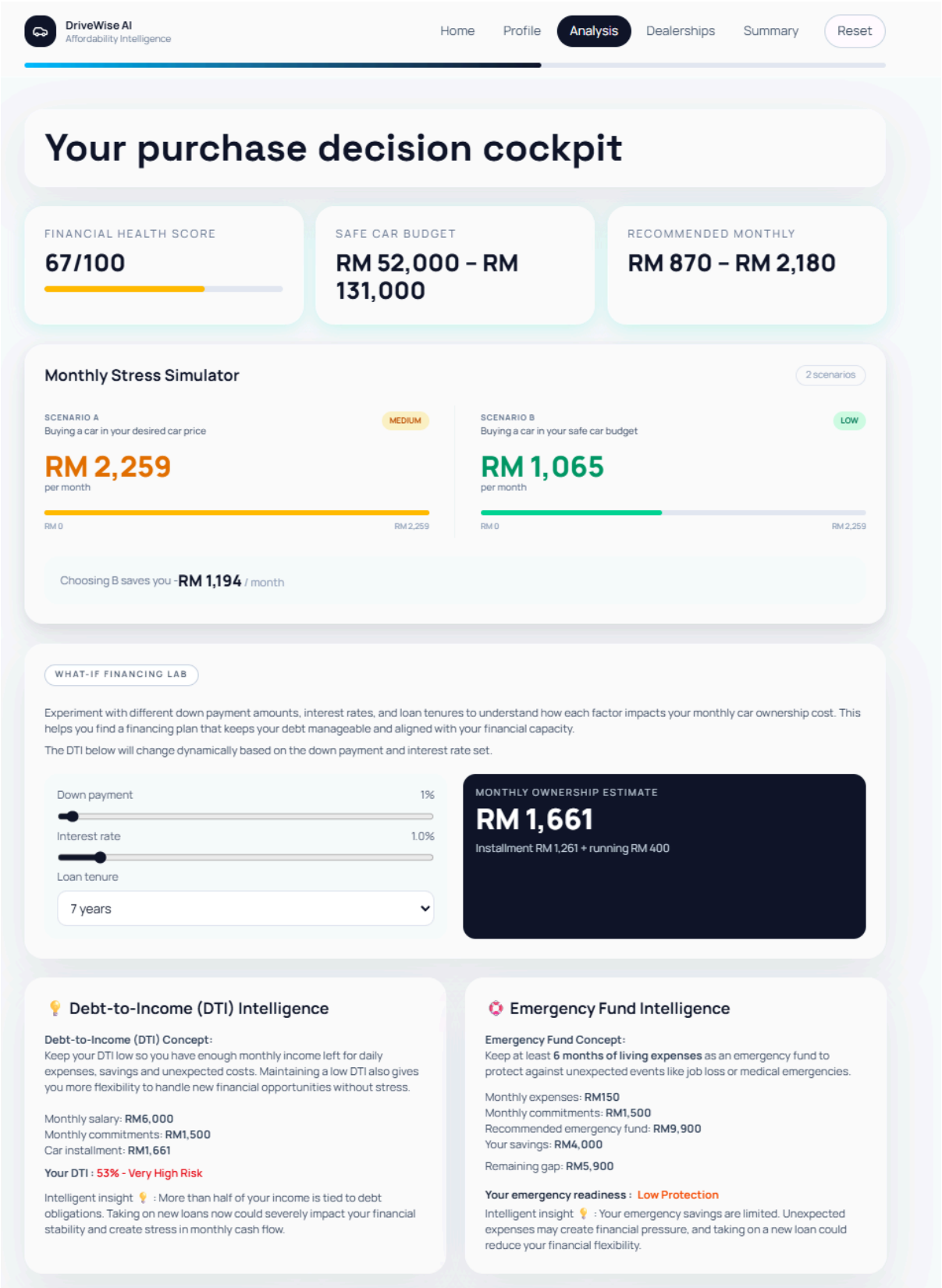
The landing page introduces the **DriveWise AI** platform and explains its purpose. Users can begin the affordability check or preview the dashboard.

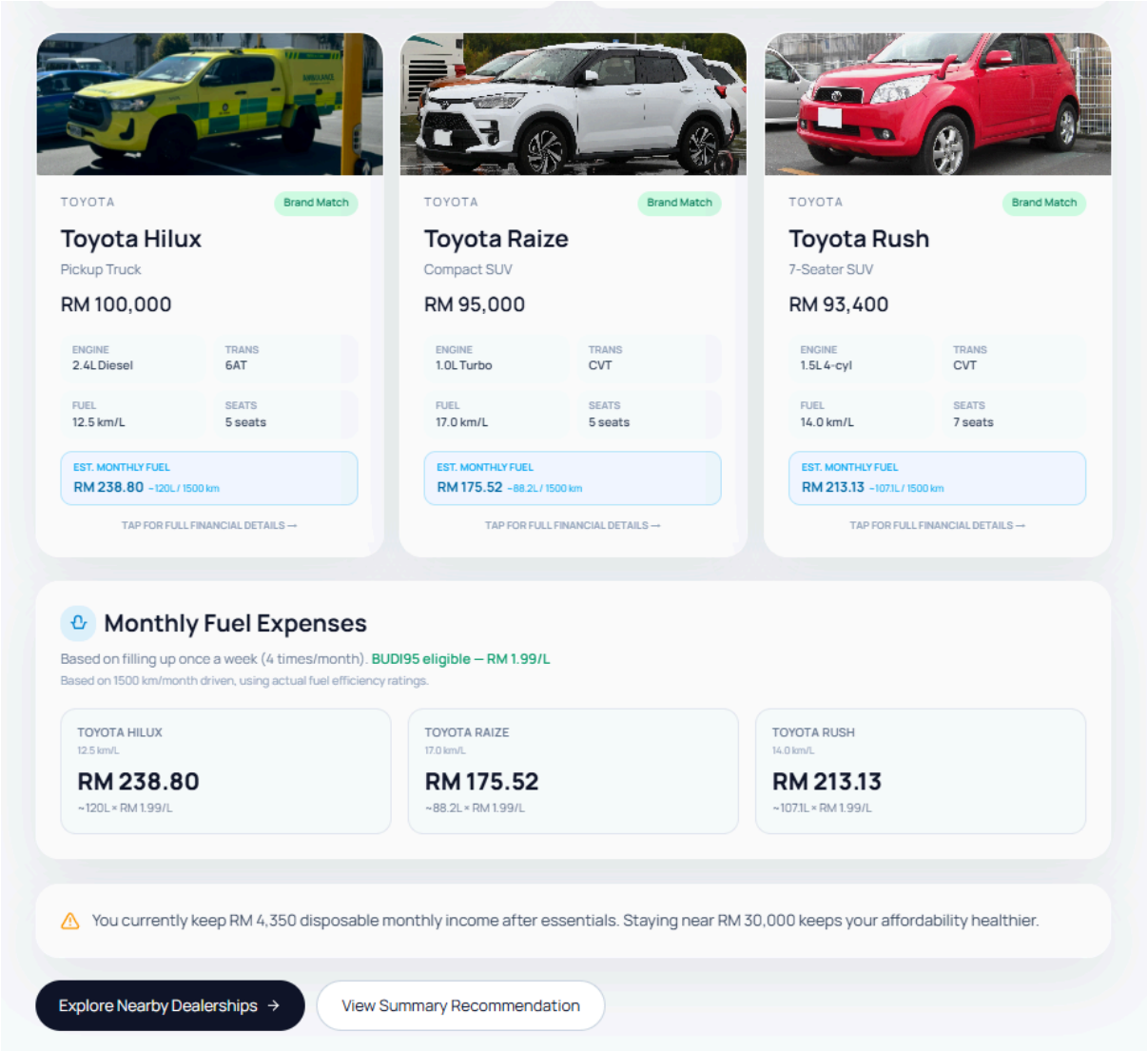
II . Financial Profile Input Page



Users provide financial information such as salary, savings, expenses, and commitments. A live preview estimates disposable income.

III . Affordability Analysis Dashboard





The dashboard visualizes financial health scores, recommended car budgets, and stress simulation scenarios, what if financing lab, recommended cars for your budget, and monthly fuel expenses based on the Budi95 eligibility

IV . Car Selection Interface

Retail price

RM 93,400

PAYMENT PER MONTH

RM 1,178

This is **27.1%** of your monthly income

ENGINE	TRANS	FUEL EFF.	SEATS
1.5L 4-cyl	CVT	14.0 km/L	7 seats

Monthly Breakdown

Loan Installment	RM 1,178
Fuel Cost ~107.1L/mo	RM 213.13
Insurance Est. comprehensive	RM 220
Road Tax 1500 cc	RM 8
Service & Maint. Est. 2×/year	RM 100
Total / Month This is **39.5%** of your monthly income	**RM 1,719.13**

Annual Breakdown

Loan Installment	RM 14,136
Fuel Cost ~107.1L/mo	RM 2,558
Insurance Est. comprehensive	RM 2,635
Road Tax 1500 cc	RM 90
Service & Maint. Est. 2×/year	RM 1,200
Total / Year	**RM 20,619**

💡 Lifestyle Impact Meter

This meter helps you understand how your current expenses, commitments, and car payments affect your monthly financial comfort.

Lifestyle : Comfortable

Intelligent Insight 💡 : Your leftover cash is healthy. You can manage unexpected costs and still have room for savings or discretionary spending, though careful planning is always good.

Breakdown:
Remaining = Salary − (Expenses + Commitments + Car Installment)
= RM6,000 − (RM150 + RM1,500 + RM1,719.13)
Remaining : RM2,630.87 (44% of your salary)

Financial Safety Score

This score evaluates your financial safety based on debt, savings, leftover cash, and loan tenure.

Status: High Risk

Intelligent Insight 💡 : A significant portion of your income is already committed to expenses and debt. Purchasing this car could strain your monthly cash flow, reduce your ability to save, and limit financial flexibility during emergencies.

📊 Breakdown:
DTI Score: 20/100 (53% DTI)
Savings Score: 40/100 (Saved RM4,000 / Recommended RM9,900)
Leftover Score: 45/100 (Remaining RM2,689 / 45%)
Loan Tenure Score: 80/100 (5 years)

Loan Details

From your What-If Financing Lab settings. Adjust them in the Analysis page.

CAR PRICE	DOWN PAYMENT	FINANCED AMOUNT
RM 93,400	RM 934 (1%)	RM 92,466
INTEREST RATE	**LOAN TENURE**	**TOTAL LOAN COST**
1.0% p.a.	7 years (84 months)	RM 98,952

⚠ Fuel cost estimated at 1500 km/month. Insurance is a first-year estimate (no NCD). Road tax based on engine displacement. Service cost is averaged across two services per year. All figures are estimates — actual costs may vary.

Life Impact Simulator

5-year savings projection based on your choice today

Your car decision today shapes your financial future. Compare three paths and see how much you could save over the next 5 years.

5-YEAR SAVINGS COMPARISON

🚗 Buy this car

RM 157,852

SAVED

RM 2,630.87/mo free cash

Hide breakdown ^

You spend a lot on this car (high monthly payment + insurance + maintenance). Because of that, you can save very little each month.

Monthly saving: RM 2,630.87
5 years = 60 months → RM 2,630.87 × 60 = RM 157,852
Result: Your total savings after 5 years = RM 157,852

🚙 RM15k Cheaper Car

RM 171,112

SAVED

+8% vs this car

Hide breakdown ^

You spend less on the car (lower monthly payment of RM 989/mo), allowing you to save more each month.

Monthly saving: RM 2,851.87
5 years → RM 2,851.87 × 60 = RM 171,112
Result: Your total savings after 5 years = RM 171,112

💰 Invest Free Cash (7%)

RM 204,174

SAVED

+29% vs this car

Hide breakdown ^

The extra money you saved from buying a cheaper car, is invested instead of spent.

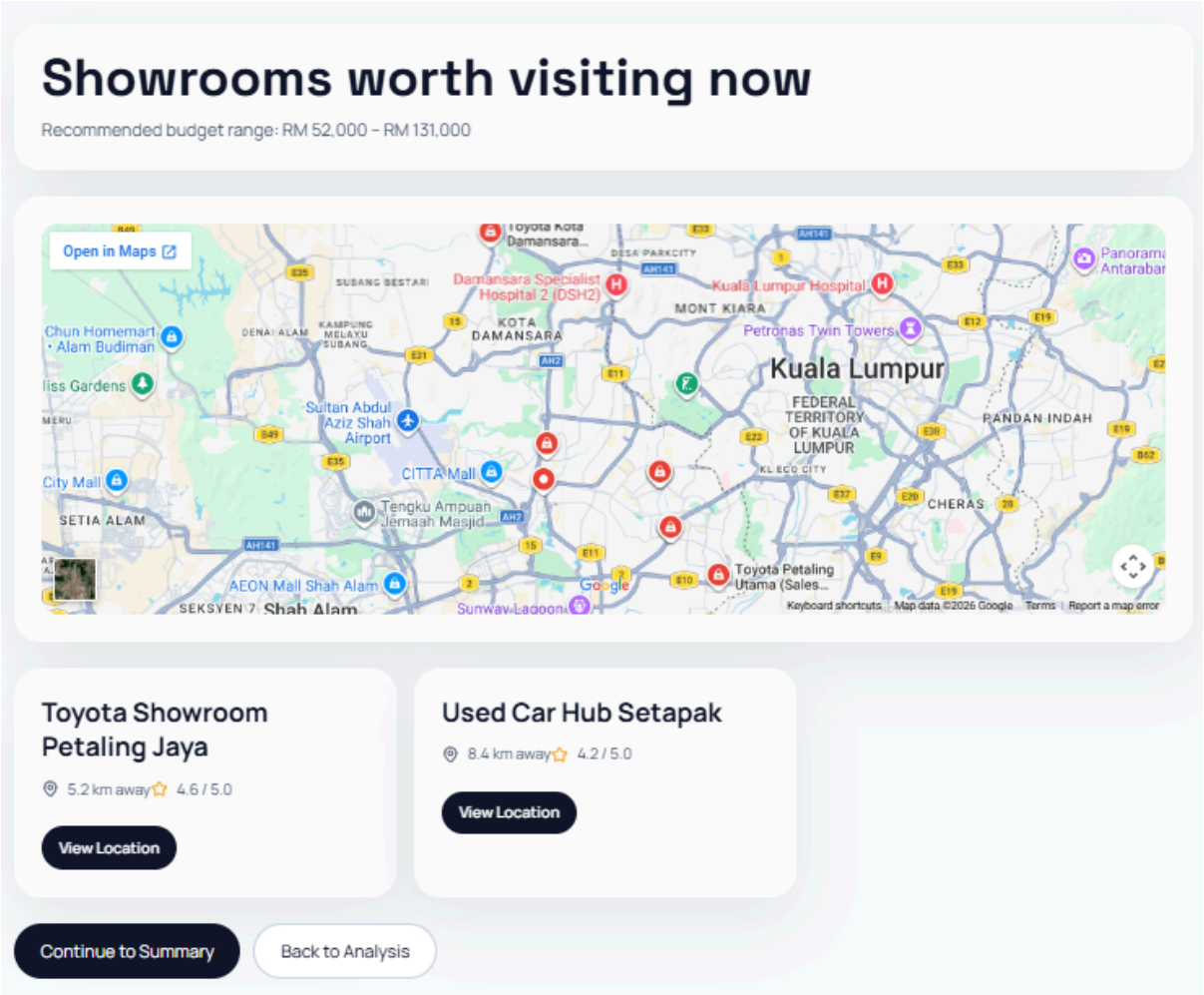
Extra money: RM 2,851.87/month
Invested at 7% p.a. growth → total = ${formatRM(li_savings_C)}
Result: Your total wealth after 5 years = RM 204,174

Figures based on your profile: RM 6,000/mo salary, RM 1,650/mo outgoings. Cheaper car scenario uses same 1% down, 1% rate, 7-year tenure. Investment return assumes 7% p.a. **Your choices today shape your financial future.**

Back to Analysis | Find Dealerships

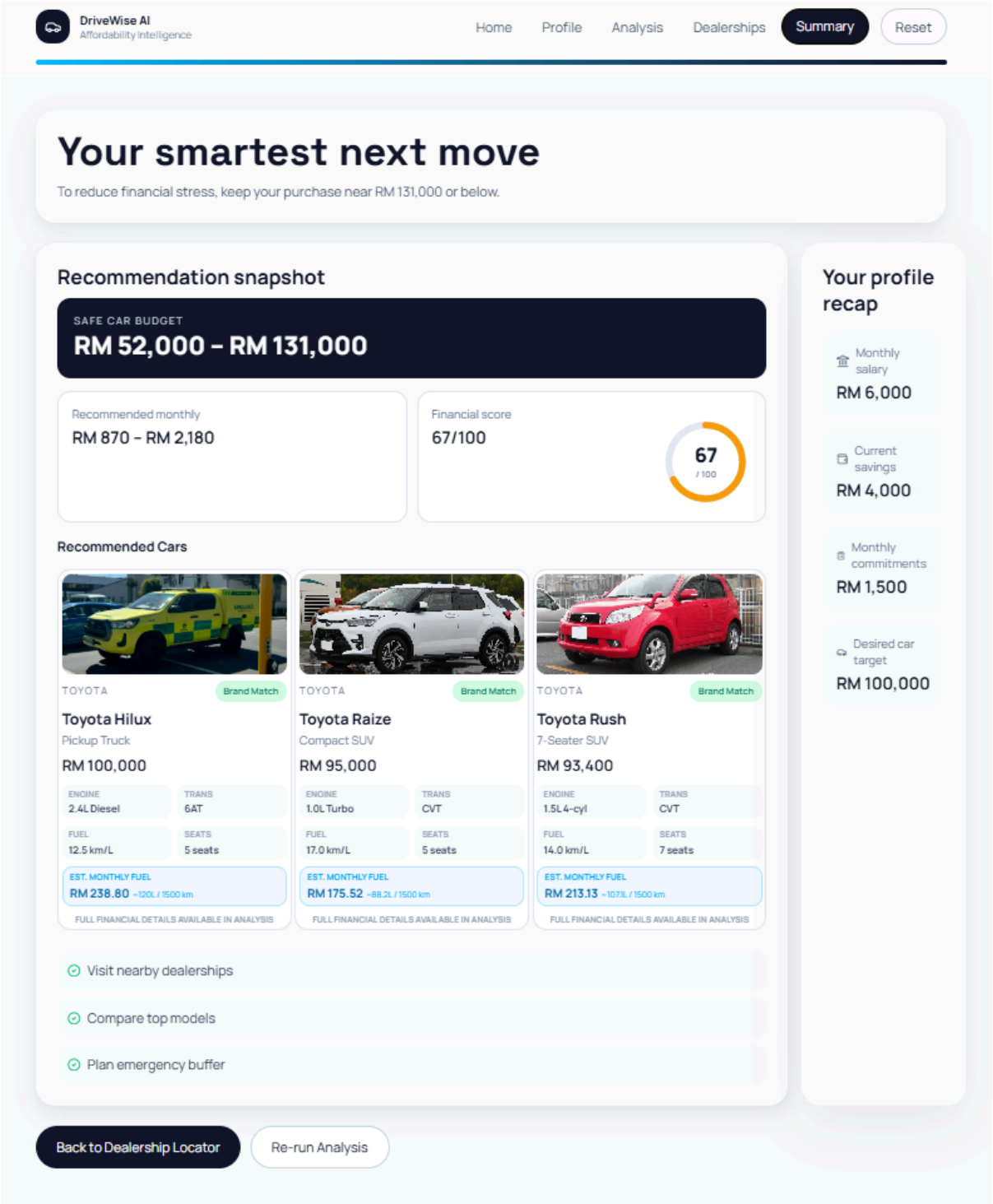
The system shows the selected car that matches the user's financial capacity and provides the specification of the car, the monthly and annual cost breakdown, Lifestyle impact Meter, Financial Safety Score and Life Impact Simulator

V . Dealership Locator Interface



Users can locate nearby dealerships where they can explore vehicles within their recommended budget

VI . DriveWise Ai Summary



Users can have the overall analysis review summary with recommended cars along with the financial scores and recommended monthly payments.

6. AI acknowledgement

Artificial Intelligence (AI) tools were utilized during the development of this project to enhance productivity and support the design and documentation process. The tools used include:

- ChatGPT – For generating ideas, drafting content, and refining written explanations.
- AI coding assistants – For assisting with frontend coding, debugging, and workflow optimization.
- Generative design tools – For visual and interface prototyping support.

These AI tools were applied to:

1. Generate initial project concepts and solution approaches.
2. Assist with frontend implementation and troubleshooting.
3. Provide guidance on system architecture and development workflow.
4. Support writing, structuring, and formatting of project documentation.

All outputs produced by AI tools were carefully reviewed, modified, and validated by the project team to ensure accuracy, relevance, and alignment with project goals.

7. Conclusion

DriveWise AI demonstrates how artificial intelligence can enhance financial decision-making for vehicle purchases. By integrating affordability simulations, financial health scoring, and dealership recommendations, the platform provides users with practical insights that reduce financial risk.

The prototype successfully showcases a user-centered solution to a common problem faced by young adults across ASEAN. Future enhancements could include integration with real-time financial data, bank loan APIs, and machine learning models for more precise affordability predictions. Expanding to mobile applications and personalized financial coaching could further increase the platform's impact.

Ultimately, **DriveWise** AI represents a step toward smarter, more responsible car purchasing decisions in the digital economy, empowering users with the tools to make informed financial choices.

8. References

- Lusardi, A., & Mitchell, O. (2014). *The Economic Importance of Financial Literacy: Theory and Evidence*. Journal of Economic Literature.

- Consumer Financial Protection Bureau. (2016). *Consumer Voices on Automobile Financing*. U.S. CFPB.

- Stango, V., & Zinman, J. (2009). *Exponential Growth Bias and Household Finance*. Journal of Finance.

- OECD. (2021). *Digital Financial Literacy – Core Competency Framework for Adults in ASEAN*. Organization for Economic Co-operation and Development.

- World Bank. (2022). *Financial Capability and Inclusion in ASEAN*. World Bank.

- Journal of Financial Literacy and Wellbeing. (2020). *Importance of Financial Literacy: Evidence from Singapore*. Cambridge University Press.

- Journal of Consumer Research. (2021). *Consumer Understanding of Auto Financing Terms*.

- Investopedia. (2023). *Car Buyers Stretching Auto Loans Longer — and It Costs More Than It Seems*.